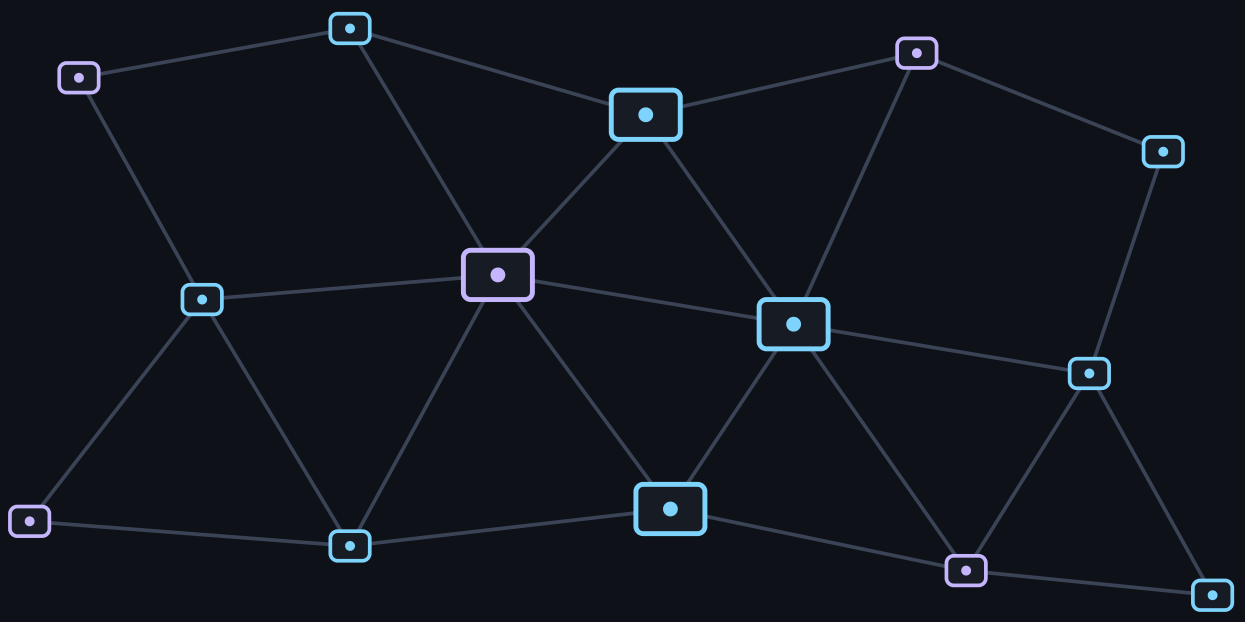
Midterm Top Questions

From Assignments 1 & 2

The 20 questions most likely to come in the midterm, ranked by priority.



13	5	2
★★★ from your list	★★ very likely	★ likely

DCS / midterm / ranking

Priority Ranking

الترتيب أول 13 سؤال (T1–T13) هما أسئلة قايمتك وكل اللي ليه علاقة بيها في الشيتين، وبعدهم الأسئلة اللي اتكررت في السكاشن. وزي ما قلت: الرسم ثم الرسم.

#	Priority	Question · why it is likely
T1	★★★	List the functions of middleware. Section 2 جه في · Middleware & Functions — من قايمتك — ASS#1 · Q11
T2	★★★	Explain and List the functions of middleware. (السؤال اتكرر في الشيت) Middleware & Functions — من قايمتك — ASS#1 · Q17
T3	★★★	Explain and List the functions of virtualization. Virtualization & Functions — من قايمتك — ASS#1 · Q19
T4	★★★	What is meant by fault tolerance challenge and how to achieve it? Fault Tolerance — من قايمتك — ASS#1 · Q16
T5	★★★	Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system? Section 1 جه في · Definition of DCS + Examples — من قايمتك — ASS#1 · Q1
T6	★★★	Compare between centralized, decentralized, and distributed computing systems? Section 1 جه في · Compare Distributed & Centralized — من قايمتك — ASS#1 · Q7
T7	★★★	Draw and explain Enslow model for distributed processing systems Section 1 جه في · Enslow Model — من قايمتك — ASS#2 · Q6
T8	★★★	Draw the DCS software architecture and compare between distributed operating system and network operating system? Compare NOS & DOS — من قايمتك — ASS#1 · Q6
T9	★★★	Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems Compare Loosely & Tightly Coupled — من قايمتك — ASS#2 · Q9
T10	★★★	Explain Tightly Coupled Multiprocessor System? (جزء من المقارنة) Tightly Coupled — من قايمتك — ASS#2 · Q7
T11	★★★	Explain Loosely Coupled Multiprocessor System? (جزء من المقارنة) Loosely Coupled — من قايمتك — ASS#2 · Q8
T12	★★★	Explain and List the functions of clustering computing. Definition of Computer Clusters — من قايمتك — ASS#1 · Q18
T13	★★★	What are the benefits of a computer cluster? Computer Clusters (benefits) — من قايمتك — ASS#1 · Q20
T14	★★	Explain what is meant by (distribution) transparency, and give examples of different types of transparency. Section 2 وفي ASS#1 جه في — ASS#1 · Q9
T15	★★	Explain the technologies that move the computer systems from centralized to distributed. Section 2 وفي ASS#2 جه في — ASS#2 · Q1
T16	★★	Identify the various challenges in the distributed system. ASS#1 اتسألت بأسئلة كتير في challenges الـ — ASS#1 · Q8
T17	★★	Explain the key components of distributed computing systems using drawing? Section 1 جه في — using drawing سؤال — ASS#1 · Q4
T18	★★	What are the types of cluster computing? cluster الـ مكمّل لتعريف — cluster أنواع الـ — ASS#1 · Q21
T19	★	Explain Multiprocessor pool distributed Systems? Cambridge مرتبط بـ processor-pool الـ — ASS#2 · Q14
T20	★	Explain workstation server distributed Multiprocessor System? رسمة سهلة — models 5 من الـ — ASS#2 · Q13

Top Questions — Answers

T1 ★★★ ● قايمتك ● متكرر

ASS#1 Q11 · Ch1 — slide 60

List the functions of middleware.

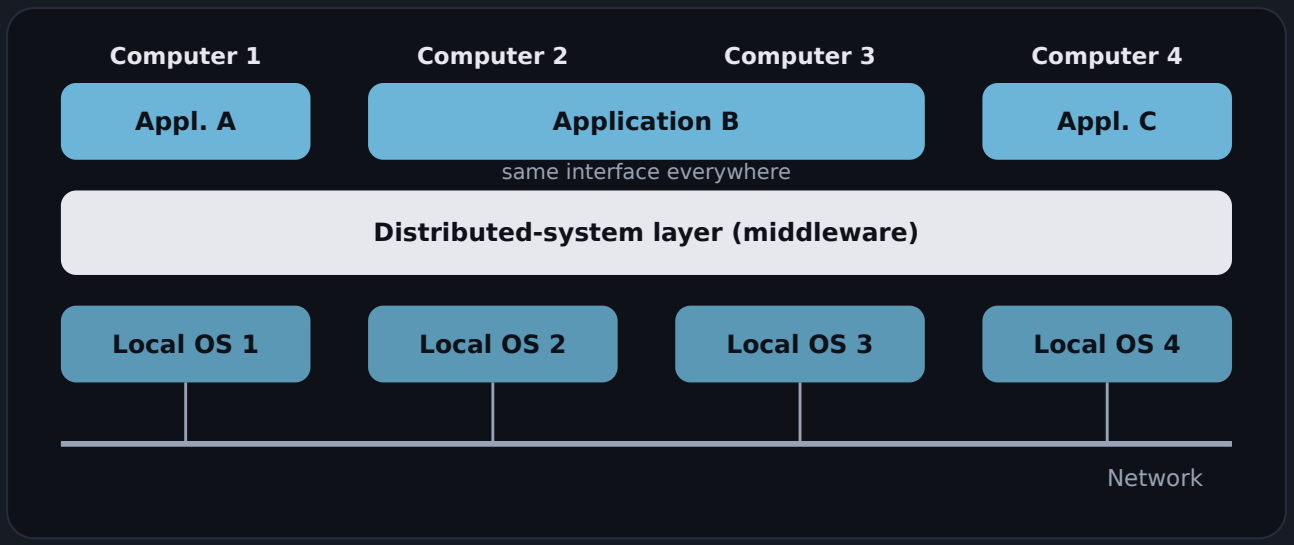


1. Layer between (DOS and NOS) and distributed applications.
2. Hides complexity and heterogeneity of the distributed system.
3. Bridges the gap between low-level OS communications and programming language abstractions.
4. Provides common programming abstraction and infrastructure for distributed applications.

الفكرة 4 وظايف — أهمهم إنه بيخبي الـ heterogeneity.

T2 ★★★ ● قايمتك ● متكرر ASS#1 Q17 · Ch1 — slides 47, 48, 60

Explain and List the functions of middleware.



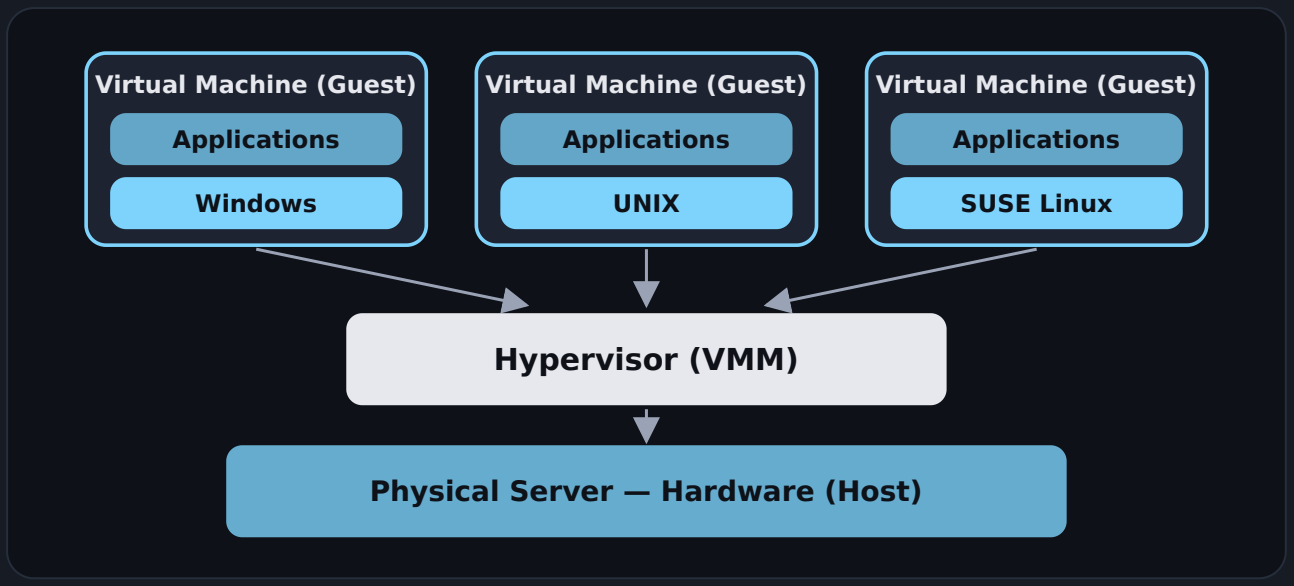
Middleware is a set of services that enables applications and end-users to interact across a heterogeneous DCS by **masking the heterogeneity** of the underlying networks, hardware, OS and programming languages; it offers each application the same interface.

1. Layer between (DOS and NOS) and distributed applications.
2. Hides complexity and heterogeneity.
3. Bridges low-level OS communications and programming language abstractions.
4. Common programming abstraction and infrastructure.

الفكرة تعريف ((masking heterogeneity + نفس الـ 4 وظايف.

T3 ★★★ ● قايمتك ASS#1 Q19 · Ch1 — slides 67–69, 71

Explain and List the functions of virtualization.



- Virtualization partitions the resources of a single platform into **multiple segregated, virtualized execution environments**, allowing multiple OSs on the same hardware.
- **Guest** = each environment · **Host** = the platform · **Hypervisor (VMM)** = the bridge between them.
- **CPU virtualization:** the hypervisor preempts an OS that consumes all the CPU. **Memory virtualization:** each OS has a virtual address space mapped to physical memory.

Functions / advantages:

1. **Multiple secure environments** (sandbox).
2. **Mixed OS environment.**
3. **Better system utilization.**
4. **Failure isolation.**

الفكرة Guest / Host / Hypervisor + 4 مميزات.

T4 ★★★ قايمتك ASS#1 Q16 · Ch1 — slide 52

What is meant by fault tolerance challenge and how to achieve it?

- Faults in hardware or software may produce incorrect results or stop before completing the computation.
- Failure handling is difficult because the failure is **partial**: some components fail while others continue to function.
- Distributed systems must maintain **availability** even at low levels of hardware/software/network reliability.

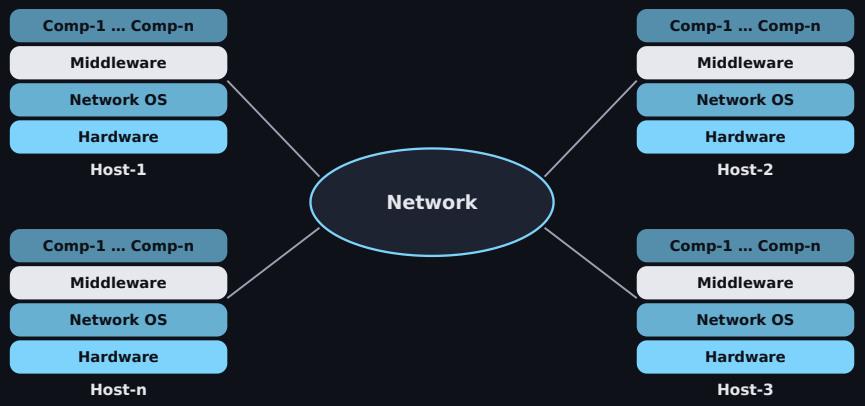
How to achieve: recovery and **redundancy**.

الفكرة Failure partial ← recovery + redundancy.

T5 ★★★ قايمتك متكرر

ASS#1 Q1 · Ch1 — slide 43 (definition) · slides 44–45 (examples) · slides 4–5 (benefits) · slide 41 (comparison)

Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?



Definition:

- A DCS is one in which components located at networked computers **communicate and coordinate their actions only by passing messages**.
- It requires **autonomous computers** interconnected through a network, each equipped with **DOS software**.
- DCS requires **transparency of distribution**: users are not aware that the system is distributed.

Pros over centralized:

1. **Scalability:** add more nodes as data grows, without a drop in performance.
2. **Fault tolerance:** if one node fails, the remaining nodes continue processing.
3. **Performance:** parallel processing of subtasks → reduced processing time.
4. **Large volumes of data:** data partitioning; each segment processed on a node.

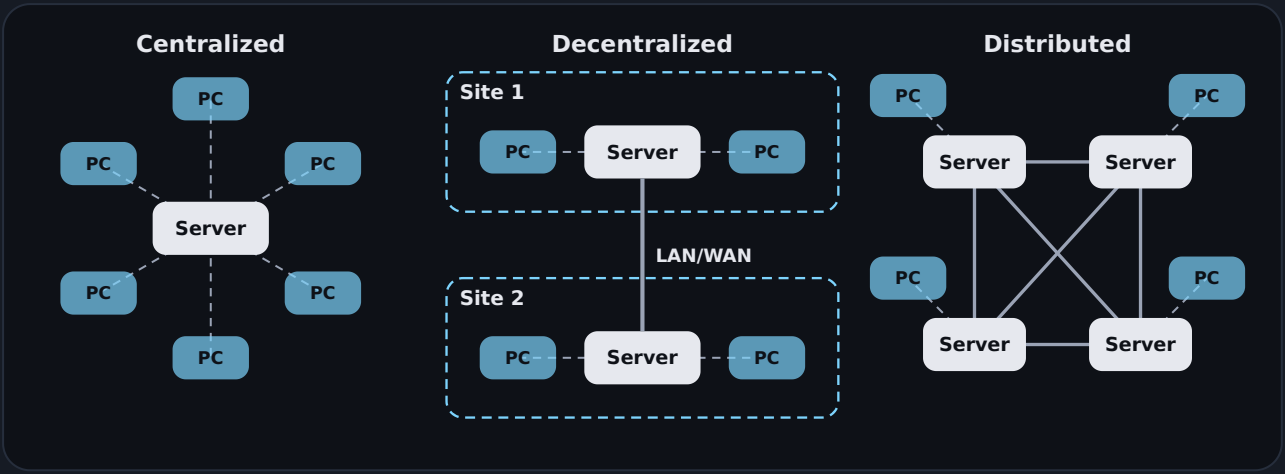
Examples (slides 43–45): Internet / WWW · Intranet and LAN · mobile and ubiquitous systems · Database Management System · ATM network · e-mail · FTP · electronic payment · real-time processing.

Cons: more complex management (coordination of many nodes) · potentially higher latency due to network communication · security is harder.

الفكرة التعريف 4 نقط + أمثلة (Internet, ATM, DBMS, LAN) + 4 مميزات.

T6 ★★★ قايمتك متكرر ASS#1 Q7 · Ch1 — slides 40, 41

Compare between centralized, decentralized, and distributed computing systems?

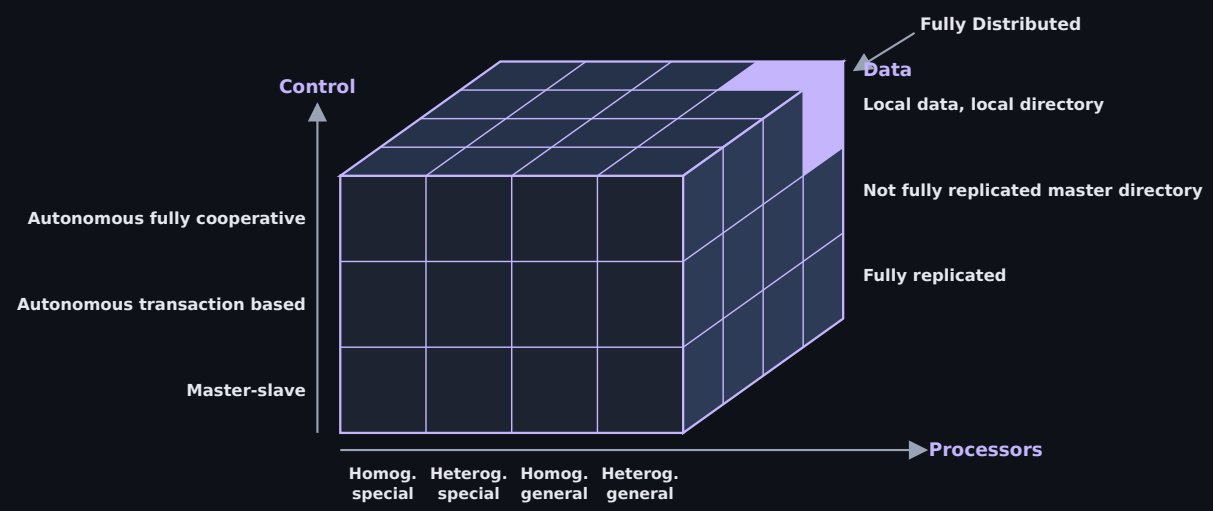


Aspect	Centralized	Decentralized	Distributed
Definition	Single central server controls all operations	Multiple nodes with independent control, no central authority	Interconnected nodes working together as a single system
Control	Single point of management	Each node operates independently	Shared; nodes collaborate
Single point of failure	High risk	Reduced risk	Reduced; fault tolerance and redundancy
Scalability	Limited, bottleneck	More scalable	Highly scalable
Resource utilization	Central server heavily utilized	Spread across nodes	Efficient sharing across nodes
Performance	Degrades with load	Improves with more nodes	High (parallel processing)
Management	Easier	More complex	Complex (coordination)
Latency	Lower	Varies with distance	Higher (network)

الفكرة احفظ الـ 8 Aspects وارسم الـ 3 رسومات.

T7 ★★★ ● قايمتك ● متكرر ASS#2 Q6 · Ch2 — slides 22–25 · Section 1 (Q4)

Draw and explain Enslow model for distributed processing systems



Distributed system = distributed hardware + distributed control + distributed data. A system is distributed if all three reach a certain degree of decentralization; **fully distributed** = fully distributed in all three.

Control:

1. Single fixed control point
2. Single dynamic control point
3. Fixed master/slave
4. Dynamic master/slave
5. Multiple homogeneous
6. Multiple heterogeneous

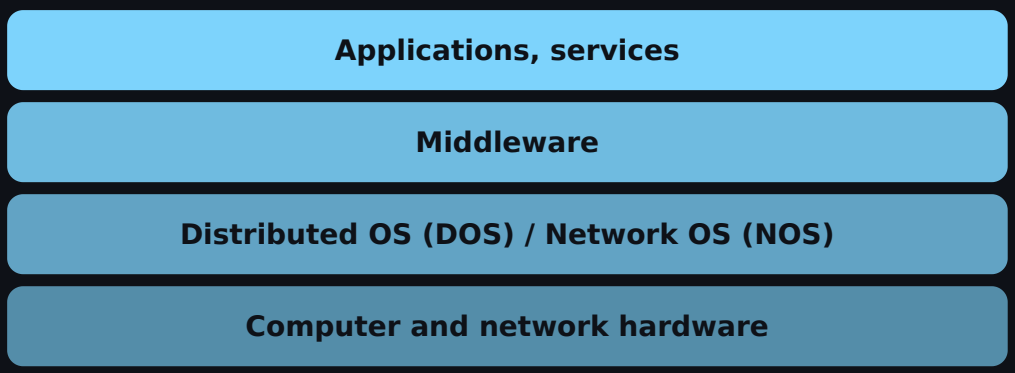
Data:

1. Centralized database
2. Distributed files, central directory
3. Replicated database
4. Partitioned, master keeps all files
5. Partitioned, master keeps directory
6. Partitioned, no master

الفكرة مكعب بـ 3 محاور (Processors, Control, Data)، والركن = fully distributed.

T8 ★★★ ● قايمتك ● متكرر ASS#1 Q6 · Ch1 — slide 57 (layers) · slide 58 (NOS vs DOS)

Draw the DCS software architecture and compare between distributed operating system and network operating system?



Network OS (NOS)	Distributed OS (DOS)
Supports networking; connects computers to share files, applications and printers	Coordinates connected nodes to operate like a **single large machine**
Each computer works self-contained, communicating through the NOS	DOS unifies all computers so they work together
MS Server, IBM AIX, Oracle Solaris, Linux Server, UNIX	MS Azure, Google File System (GFS), LOCUS
Resource sharing · centralized control · scalability · NOS security	Resource utilization · transparency · fault tolerance · scalability · DOS security

الفكرة كل layers: Applications ← Middleware ← DOS/NOS ← Hardware. NOS 4 جهاز لوحده، DOS الكل جهاز واحد.

T9 ★★★ قايمتك ASS#2 Q9 · Ch2 — slide 30

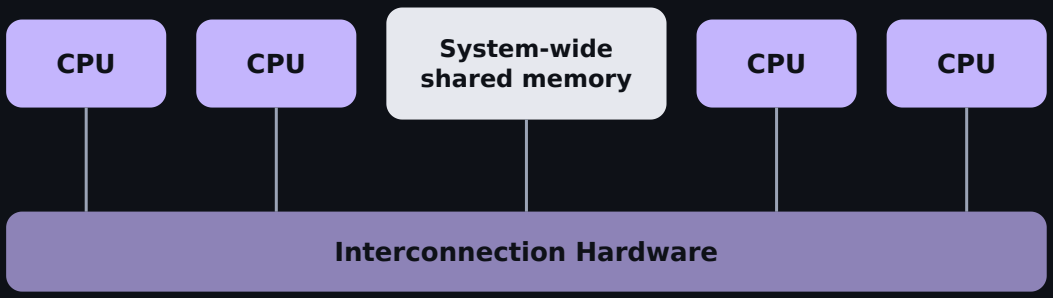
Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems

Loosely Coupled	Tightly Coupled
Distributed memory	Shared memory
Low data rate	High data rate
Memory conflicts don’t take place	Memory conflicts
Low degree of interaction between tasks	High degree of interaction between tasks
Direct connection between processor and I/O devices	IOPIN connects processor and I/O devices
Applications: DCSs	Applications: parallel processing

الفكرة كل سطر عكس اللي قصاده.

T10 ★★★ قايمتك ASS#2 Q7 · Ch2 — slide 28

Explain Tightly Coupled Multiprocessor System?

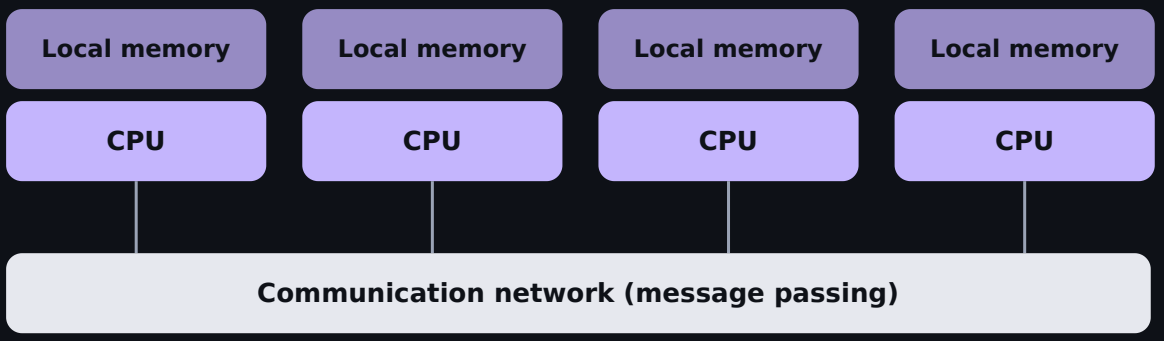


- Tightly coupled systems are **multiprocessor systems** sharing addressing to memory and devices using a **single OS**.
- Called parallel processing or **shared memory** systems: one system-wide primary memory shared by all processors.
- If a processor writes 100 to location x, any other processor reading x gets 100.
- The number of processors is small, limited by the bandwidth of the shared memory.

الفكرة Shared memory + single OS.

T11 ★★★ قايمتك ASS#2 Q8 · Ch2 — slide 29

Explain Loosely Coupled Multiprocessor System?

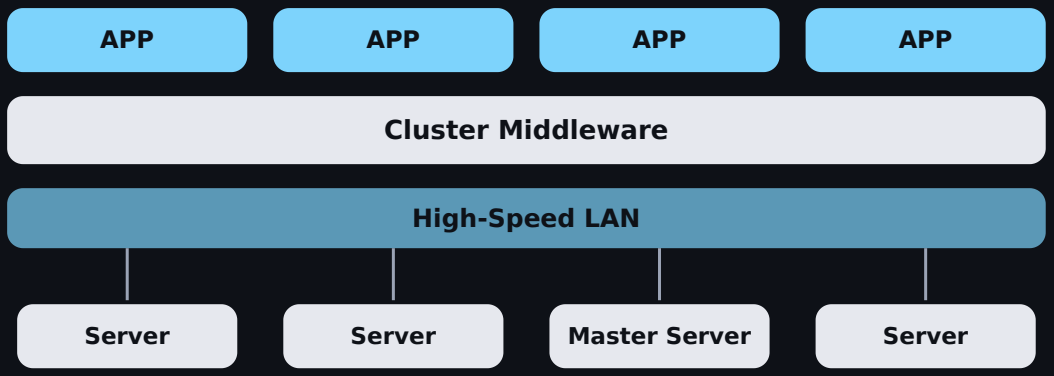


- Comes very close to distributed processing: **multiple complete systems**.
- Connected over I/O pathways; they **do not share memory or operating systems**.
- They may be functionally dependent and act as a single node.

الفكرة Private memory، مفيش shared memory ولا OS.

T12 ★★★ ● قايمتك ASS#1 Q18 · Ch1 — slides 62, 64

Explain and List the functions of clustering computing.



- A cluster is a group of linked computers working together so closely that they form a **single computer**, connected through **fast LANs**.
- A large task is broken into discrete parts distributed across computers and processed simultaneously.

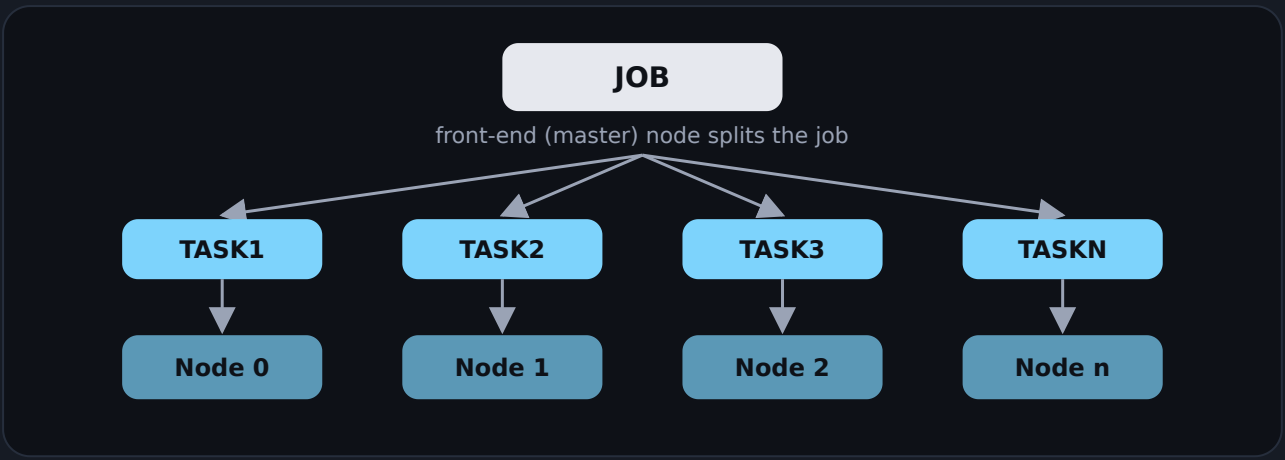
Functions: improve **performance and/or availability** more cost-effectively than one computer · parallel processing · redundancy (failover) · load balancing.

Cluster consists of: nodes (master + computing) · network · OSs · cluster middleware.

الفكرة Cluster = كذا كمبيوتر على LAN سريعة كأنهم كمبيوتر واحد.

T13 ★★★ ● قايمتك ASS#1 Q20 · Ch1 — slides 62, 66

What are the benefits of a computer cluster?



1. **Performance:** a job is split into tasks processed simultaneously → higher speed.
2. **Availability:** redundancy; no single point of failure.
3. **Cost-effective:** cheaper than a single computer of comparable speed or availability.
4. **Scalable parallel and distributed computing** in science and business.
5. **Cloud services:** high-performance computing, mass storage (data centers), web search.

الفكرة Performance + Availability + Cost + Cloud services.

T14 ★★ متكرر ASS#1 Q9 · Ch1 — slides 54, 55 · Section 2 (Q1)

Explain what is meant by (distribution) transparency, and give examples of different types of transparency.

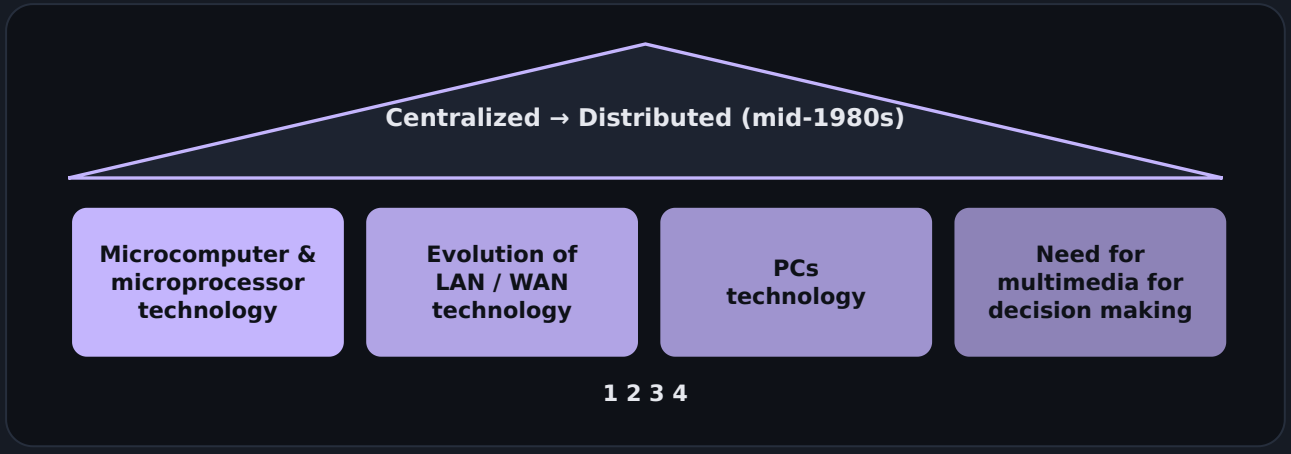
Transparency ensures that the DCS is perceived as a **single entity** by the users or the application programmers rather than a collection of cooperating autonomous systems; the user is unaware where services are located.

Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource is replicated
Concurrency	Hide that a resource may be shared by several competitive users
Failure	Hide the failure and recovery of a resource
Parallelism	Automatic use of parallelism without programming it explicitly

الفكرة 8 أنواع، كل نوع بيخبي حاجة واحدة.

T15 ★★ ● متكرر ASS#2 Q1 · Ch2 — slides 3, 15 · Section 2 (Q3)

Explain the technologies that move the computer systems from centralized to distributed.



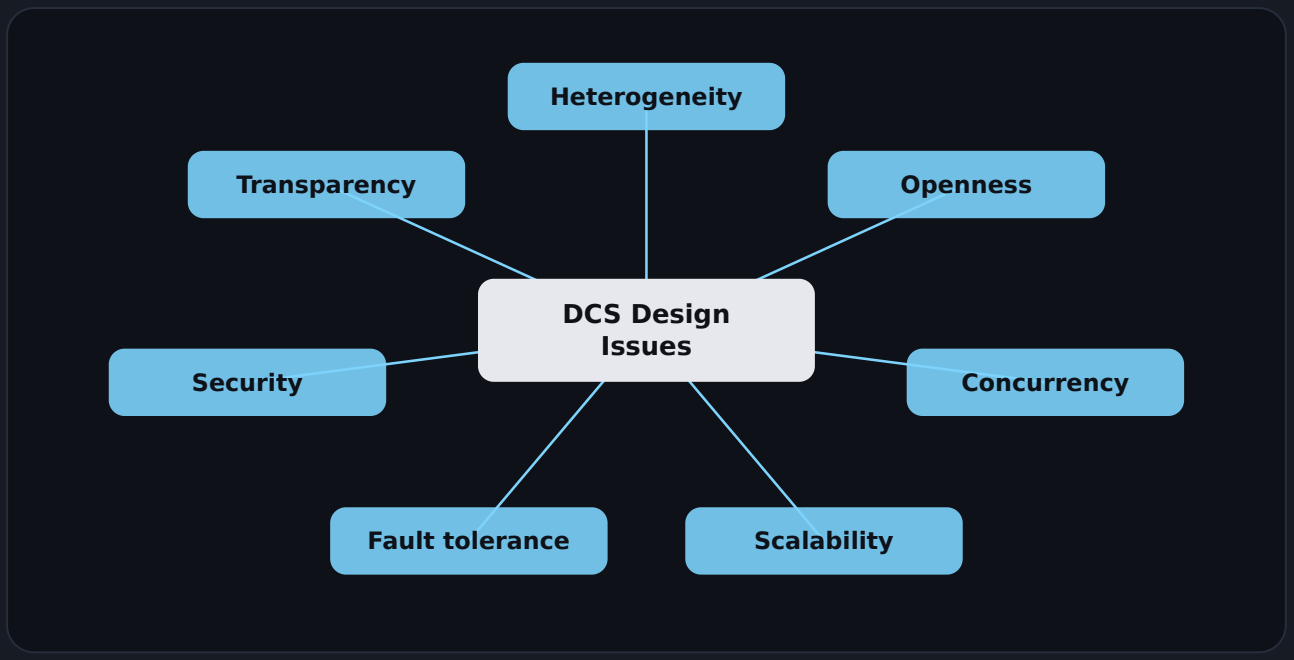
1. **Microcomputer and microprocessor technology:** 8 → 64-bit and multicore CPUs; mainframe power for 1/1000th of the price.
2. **Evolution of LAN technology:** high-speed LANs and WANs connect thousands to millions of machines.
3. **PCs technology:** miniaturization — smartphones, plug and nano computers.
4. **Need for multimedia for decision making:** PCs and powerful servers replace text terminals.

Feature	Notes
Technological changes	VLSI favors multiple low-performance processors; communication costs fell; PCs replace terminals.
User needs	Computing power where needed; managers need multimedia data for decisions.

الفكرة 4 تكنولوجيات: Microprocessor, LAN/WAN, PCs, Multimedia.

T16 ★★ ● متكرر ASS#1 Q8 · Ch1 — slide 46 · Ch2 — slides 39–42

Identify the various challenges in the distributed system.



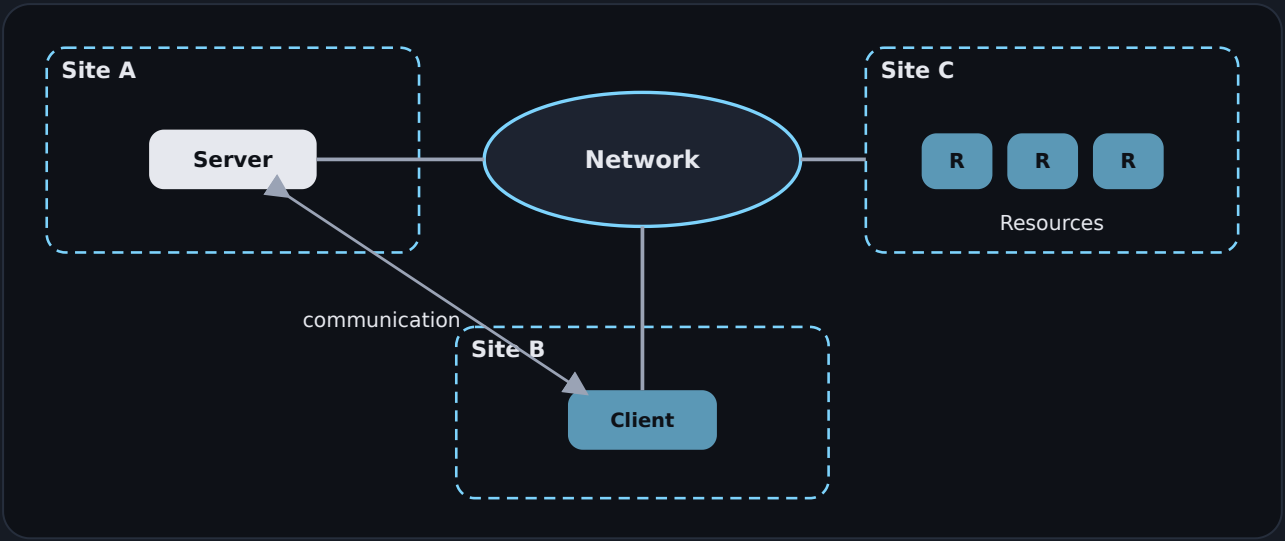
1. **Heterogeneity:** networks, hardware, OS, languages and developers differ → middleware.
2. **Openness:** adding new resource-sharing services → published interfaces.
3. **Security:** confidentiality, integrity, availability (CIA).
4. **Scalability:** remain efficient as users and resources grow.
5. **Failure handling:** partial failures → recovery and redundancy.
6. **Concurrency:** safe simultaneous access to shared resources.
7. **Transparency:** perceived as a single system.
8. **Resource sharing:** controlled sharing of hardware, data and services.

الفكرة 8 تحديات، كل واحد وجنبه حله.

T17 ★★ ● متكرر

ASS#1 Q4 · Ch1 — slide 38 · drawing: slide 89

Explain the key components of distributed computing systems using drawing?



1. **Geographical Distribution:** nodes spread across different locations, communicating via LAN or the Internet.
2. **Resource Sharing:** nodes share processing power, storage, and data.
3. **Concurrency:** multiple nodes operate concurrently → better performance and throughput.
4. **Scalability:** easy to scale by adding more nodes.
5. **Fault Tolerance:** redundancy and replication keep the system operational if some nodes fail.
6. **Transparency:** complexity is hidden; users see a single entity.

الفكرة 6 خصائص + ارسم Site C – Site B (client) – Network – Site A (server) (resources).

T18 ★★

ASS#1 Q21 · Ch1 — slide 63

What are the types of cluster computing?

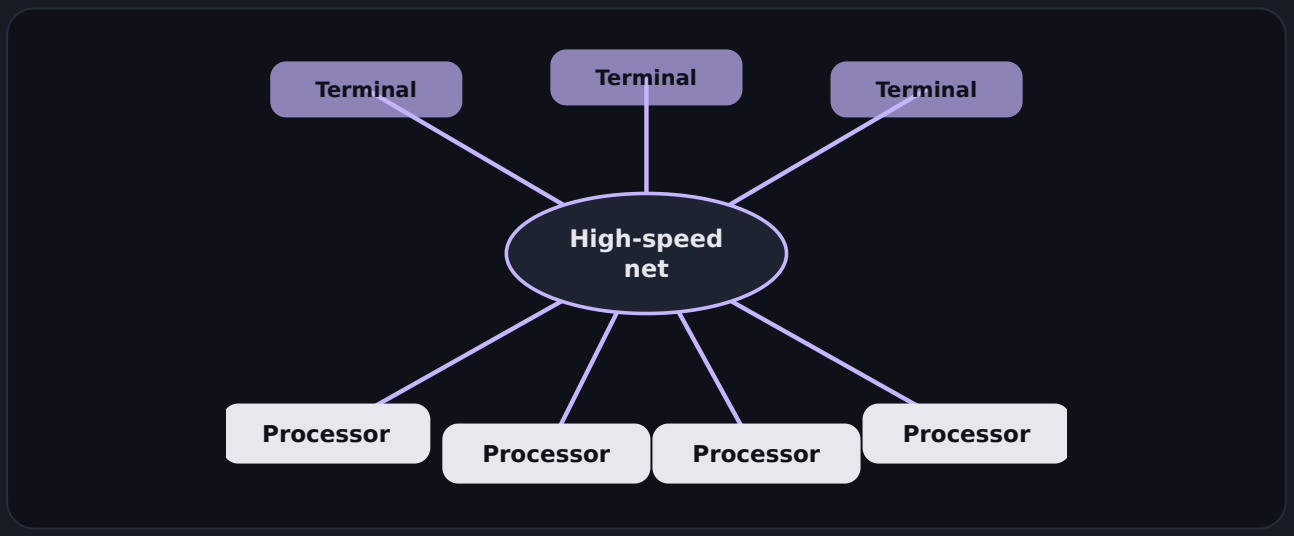
1. **High availability (HA) clusters (Linux):** for mission-critical applications; failover clusters provide redundancy and **eliminate single point of failure**.
2. **Network load balancing clusters:** distribute a workload over multiple backend nodes with redundant load-balancing front ends; all servers (web, mail, FTP) process requests.
3. **Science clusters:** a **master** computer breaks the problem into pieces and sends a piece to each **node**, sending more as nodes finish → cheap powerful supercomputer.

الفكرة 3 أنواع: HA – Load balancing – Science.

T19 ★

ASS#2 Q14 · Ch2 — slide 37

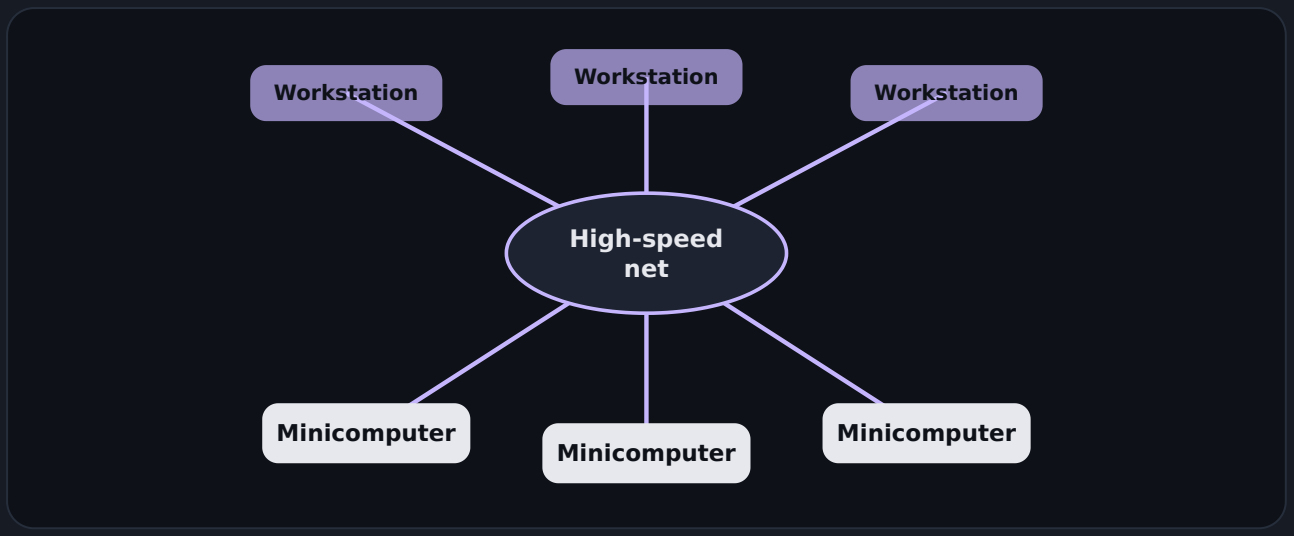
Explain Multiprocessor pool distributed Systems?



- Most of the time a user needs no computing power, but once in a while needs a **very large amount** for a short time.
- Processors are **pooled** and shared by users as needed; each has its own memory.
- No home machine; the user does not log onto any machine → better utilization and flexibility. Example: **Cambridge DCS**.

الفكرة Pool مشترك + مفيش home machine.

Explain workstation server distributed Multiprocessor System?



- Multiple workstations coupled with **powerful servers** (file systems, databases).
- A user logs onto his own workstation; normal computation is done locally.
- Requests for special services are sent to a **server** that performs them and returns the result; processes need not migrate to the server.

الفكرة الشغل العادي local، والخدمات الخاصة على الـ server.